Dynamic Common and Unique Feature Fusion Network for Hyperspectral and LiDAR Data Classification

Changzhe Jiao, *Member, IEEE*, Lei Wang, Chao Hu, Xu Tang, *Senior Member, IEEE*, Hao Zhu, *Member, IEEE*, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—With the advancement of multimodal technology, the combination of hyperspectral image (HSI) and light detection and ranging (LiDAR) data has become a prominent research topic in land use and land cover (LULC) classification. There are complex correlations between HSI and LiDAR. However, many studies focus on the spectral, spatial, and other attribute features of multimodal data while ignoring these important relationships, thereby limiting the classification performance. To address these, we propose a dynamic common and unique feature fusion network (DCU-Net) that establishes the dependence relationship between HSI and LiDAR and mines their shared and complementary information. A multiscale attribute feature extraction block is employed to capture spectral–spatial information of HSI and spatial–elevation information of LiDAR data, which effectively reduces the effect of scale differences among objects. In addition, we introduce a novel common-unique transformer (CUT) block with cross dynamic-agent-attention to extract common features of HSI and LiDAR data and depthwise convolution modules focusing on their unique features. By associating common and unique features of HSI and LiDAR with their dependencies, the robustness and classification accuracy of the model are significantly improved. In the fusion stage, common and unique features are adaptively reconstructed into discriminative features containing high-level semantic information. Extensive experiments on four popular HSI and LiDAR datasets illustrate the superiority and effectiveness of our proposed model, which showcases the great potential for multimodal remote sensing (RS) data analysis. The source code of the proposed method is available publicly at https://github.com/wanglei1588/Comon_Unique**

***Index Terms*—Classification, fusion, hyperspectral image (HSI), light detection and ranging (LiDAR) data.**

I. Introduction

LAND use and land cover (LULC) classification is a technology aiming at assigning a unique label to each pixel in a remote sensing (RS) image based on its ground cover information. It is one of the most important and challenging Earth observation tasks, which is widely used in urban planning [1], environmental monitoring [2], space exploration [3], and others [4], [5]. Hyperspectral image (HSI) provides detailed spectral information by sampling the reflective portion of the electromagnetic spectrum, covering a wide range from 0.4 to 2.4 μm, and HSI also contains spatial domain information [6]. HSI has been extensively used in the task of LULC classification and has obtained good classification results [7], [8]. However, in complex and dense urban areas, there are often many targets with similar spectral responses [9]. The accuracy and interpretability of LULC classification based solely on HSI obtained from a single sensor are often insufficient [10].

Thanks to the rapid development of aerospace and RS technology, people can now use multiple sensors to effectively capture multisource and multimodal RS data for the same scene on the Earth's surface, including HSI, multispectral image (MSI), synthetic aperture radar (SAR), and light detection and ranging (LiDAR) data [11]. These multimodal RS data can reflect different urban structures and material characteristics, aiding in tasks related to comprehensive scene understanding [12]. A large number of studies have proven that utilizing complementary information from data of different modalities can improve the performance of LULC classification [13], [14], [15]. HSI can effectively distinguish objects of different materials in the scene [16], but cannot distinguish objects of different heights. LiDAR reflects the elevation distribution of the detection area and can capture the vertical structural information of ground objects, which can effectively distinguish ground objects of different heights, but cannot distinguish objects of different materials. Therefore, combining HSI and LiDAR data has attracted widespread attention and has been widely applied in LULC classification.

Early works on LULC classification mostly relied on morphological operators and machine-learning algorithms. Melgani and Bruzzone [17] demonstrated the superiority of support vector machines (SVMs) over traditional feature reduction methods by evaluating their performance across HSI subspaces of varying dimensions. Ham et al. [18] investigated the use of random forest (RF) classifiers within a binary hierarchical multiclassifier framework, aiming to enhance the generalization capabilities of the classifier in HSI analysis. Li et al. [19] employed local binary patterns (LBPs) to capture local RS image features, such as corners, edges, and spots, and

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of the Education Ministry of China, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: haozhu@xidian.edu.cn).

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subsequently employed the extreme learning machine (ELM) with a straightforward architecture for classification. Machine-learning algorithms can automatically learn high-dimensional complex features in RS images by training on large amounts of data. However, these methods make insufficient use of spatial information, and parameter tuning is challenging, resulting in unsatisfied generalization and stability. In [20], information captured from HSI and LiDAR data was integrated using morphologically extended attribute profiles, facilitating the combination of spatial, spectral, and elevation features. In [21], a generalized graph-based fusion method was proposed to couple dimension reduction and feature fusion of the spectral information of HSI and morphological profiles built on both HSI and LiDAR data. Rasti et al. [22] applied extinction profiles to automatically extract spatial and elevation information from HSI and rasterized LiDAR features. Morphological operators can extract spatial structural features such as edges and shapes from RS images and effectively suppress redundant noise. However, these methods struggle to extract nonlinear spectral features and rely on the selection of structural elements, making it difficult to distinguish the class of objects in complex scenes.

With the rapid development of deep learning (DL), many works based on DL have achieved great success in LULC classification. Among them, the convolutional neural network (CNN) and the recurrent neural network (RNN) are two widely used models. CNN can automatically learn and extract spatial detail information in images through their powerful nonlinear expression ability and hierarchical feature extraction mechanism. Hu et al. [23] extracted spectral features of HSI using a 1-D CNN with five convolutional layers for each pixel, achieving better classification performance than traditional machine learning methods. Zhao and Du [24] proposed a classification framework that jointly uses dimension reduction and 2-D CNNs to capture spectral and spatial features of HSI. In [25], a 3-D CNN-based model with combined regularization was proposed to extract effective deep features of HSI that were nonlinear, invariant, and discriminant. However, CNNs may not perform well in processing sequence data and long-distance dependencies. RNNs, especially long short-term memory (LSTM) network, can capture spectral dependencies in HSI more effectively [26]. Wu and Prasad [27] proposed a convolutional RNN composed of several convolutional and recurrent layers to learn spectral and spatial features of HSI. Song et al. [28] used LSTM to extract spatial–spectral features in six directions, and then applied an attention-based fusion module to integrate these features. To better preserve the intrinsic structure information of HSI, Shi et al. [29] proposed a 3-D ConvLSTM neural network to capture spatial–spectral information. In addition, effective data fusion and information interaction methods can improve the accuracy of LULC classification. Xu et al. [30] proposed a dual-channel CNN to extract the spectral–spatial features of HSI and the elevation features of LiDAR, respectively, and then fused these features in the cascade network. Hang et al. [31] used two coupled CNNs with shared parameters to extract features of HSI and LiDAR data, respectively. They then simultaneously employed feature-level and decision-level fusion methods to integrate these heterogeneous features. In [32], a modality attention module was proposed to generate enhanced features. By considering the correlation and heterogeneity between HSI and LiDAR data, feature interaction and integration were achieved. These existing studies mainly focus on the spectral, spatial, elevation, and other attribute features of HSI and LiDAR, while ignoring the complex and important relationships between multimodal data, thereby limiting the classification performance.

In addition to attribute features, multimodal RS data also contains common features, which jointly represent land cover classes, and unique features, which provide supplementary descriptions of land cover attributes. Common features capture shared information across multimodal data, while unique features represent complementary information specific to each modality. Attribute features emphasize the physical significance within a single modality. In contrast, common and unique features highlight the correlations and complementarities between modalities, offering a higher level modeling of attribute features. In this article, we propose a dynamic common and unique feature fusion network (DCU-Net) to establish the dependence relationship between HSI and LiDAR and mine their shared and complementary information. As shown in Fig. 1, extensive experiments on four popular HSI and LiDAR datasets illustrate the superiority of our proposed model. The main contributions of this article can be summarized as follows.

1) We explicitly decompose the common and unique features of HSI and LiDAR data by the proposed DCU-Net, which allows for better utilization of shared information and complementary information while establishing dependencies between HSI and LiDAR data.
2) We propose a novel quaternary cross dynamic agent attention (cross-DAA) mechanism that consists of two softmax attention operations. The attention dynamically and progressively interacts between HSI and LiDAR to capture refined common features, reducing inconsistency and redundancy between them.
3) The proposed feature fusion block reconstructs the common and unique features of HSI and LiDAR into discriminative features that contain high-level semantic information, providing a richer and more comprehensive feature representation for complex objects.

The remainder of this article is organized as follows. Section II describes the related works. The framework of the proposed model is introduced in Section III. Section IV details the experiments, including descriptions of the datasets and implementation, evaluation of both performance and parameters, comparison results and analysis, and ablation studies. Finally, some concluding remarks are summarized in Section V.

II. Related Work

A. Attention in LULC Classification

The multihead self-attention (MSA) module is the key component in transformer [33], which has achieved great success in natural language processing. Since the inception of vision transformer [34], MSA has shown huge potential

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in the realm of computer vision and plays an increasingly important role in computer vision. The MSA computes the similarity between queries and each key-value pair, resulting in significant computational overhead. To end this, many works attempt to reduce computation complexity by improving the attention mechanism. PVT [35] employed a sparse attention [36] pattern and Swin-transformer [37] and limited attention calculation to local windows. Agent attention [38] introduced a small group of agent tokens on the basis of MSA to aggregate and broadcast global information, integrating Softmax and linear attention. In addition, researchers have proposed channel attention, such as SENet [39] and ECANet [40], which implicitly and adaptively learned the potential key features.

As an effective tool, attention learning has proven to be valuable in enhancing target features and improving the overall performance of LULC classification. Hong et al. [41] introduced SpectralFormer, which captured spectrally local sequence information by leveraging neighboring bands in HSI. To obtain deep semantic features, Sun et al. [42] proposed a spectral–spatial feature tokenization transformer (SSFTT), a method that employed a transformer encoder to model local spatial contextual information and represented relationships between neighboring sequences. These methods perform well with HSI data; however, they cannot use multimodal RS data for joint classification. In order to build an interaction between multimodal data, Ding et al. [43] proposed GLT-Net to explore local and global information from HSI and LiDAR data. Zhao et al. [44] proposed a joint classification network that incorporates a hierarchical CNN and a transformer to utilize the complementary information in HSI and LiDAR data. Roy et al. [45] introduced a multimodal fusion transformer (MFT) network, which used cross-patch attention for HSI and LiDAR data classification. In [46], an extended vision transformer (ExViT) was proposed to process multimodal RS image patches through parallel branches of position-shared ViTs extended with separable convolution modules. However, the above-mentioned approaches are restricted to information exchange between multimodal RS data, ignoring their unique features.

B. Common and Unique Feature in Multimodal Image

Multimodal image processing has consistently been a critical and challenging issue in computer vision. To address general multimodal image restoration and image fusion problems, Deng and Dragotti [47] introduced a multimodal convolutional sparse coding model to split common and unique features from multimodal images. In [48], two pairs of encoder–decoder networks were adopted to implement common and unique feature extraction and decomposition for visible and infrared image fusion. Zhao et al. [49] introduced a dual-branch transformer-CNN feature extractor, where lite transformer blocks were used to handle low-frequency global features and invertible neural network blocks were employed to extract high-frequency local information. They treat low-frequency information in multimodal images as common features and treat high-frequency information as unique features.

We can conclude that common and unique features are crucial in multimodal image processing. In this work, we extract common and unique features from HSI and LiDAR data by a dynamic approach for LULC classification.

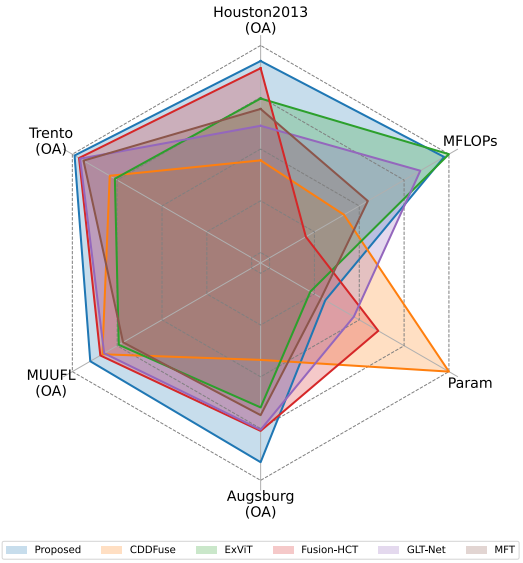


Fig. 1. Radar chart of experimental results of our proposed model and cutting-edge methods in terms of classification performance metric (OA), parameters, and computational cost (MFLOPs) on four datasets. According to the chart, our proposed model significantly outperforms its competitors along most metrics, showcasing its great potential as a novel, efficient, and effective DL framework for HSI and LiDAR multimodal data analysis.

III. Methodology

The overall architecture of our proposed DCU-Net is shown in Fig. 2. The inputs $X_H \in \mathbb{R}^{p\times p\times c}$ and $X_L \in \mathbb{R}^{p\times p\times l}$ are image patches centered on the same target pixel of HSI and LiDAR data, respectively, where p is the spatial size of patches, c and l are the channel numbers of HSI and LiDAR. Then, input data are fed into the multiscale attribute feature extraction block, i.e., the CUT block, and the common and unique feature fusion (CUFF) block. Finally, the fused features are fed into the classifier to obtain the classification result.

A. Multiscale Attribute Feature Extraction Block

HSI mainly includes spectral and spatial information, while LiDAR mainly encompasses elevation and spatial information. Multiscale features capture information at different resolutions, providing a more comprehensive description of terrain. Integrating multiscale features reduces the impact of noise, lighting changes, and occlusion on model performance, thereby improving the robustness of the model. As shown in Fig. 2, the proposed global–local attribute feature extraction (GLAFE) block extracts global–local attribute features from HSI and LiDAR data, which consists of attribute features extraction (AFE) module, multiscale convolution (MSC) module, and dynamically tokenize (DT) module.

1) AFE Module: CNN and LSTM are widely used to extract attribute features of HSI and LiDAR in recent works. While CNN effectively extracts spatial features, it destroys the spectral dependence and ignores the spectral features of HSI. Conversely, LSTM is proficient at capturing spectral features but neglects spatial features of HSI. To extract spatial

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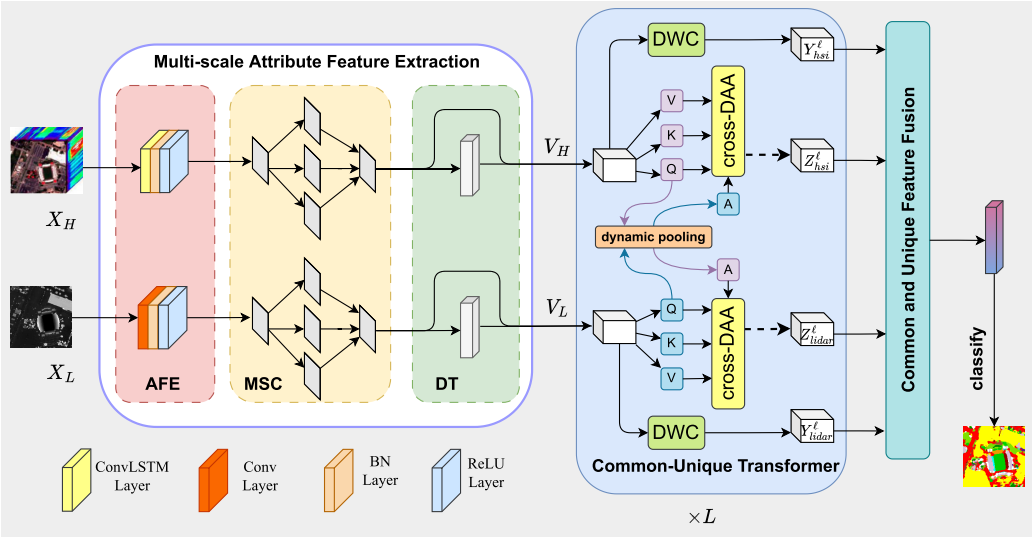


Fig. 2. Overall architecture of proposed DCU-Net, which consists of GLAFE block, CUT, and CUFF block.

features and retain the long-range spectral dependence of HSI, we develop a multimodal 3-D ConvLSTM layer to extract the spatial–spectral features of HSI. In addition, a 2-D convolutional layer is employed to extract the elevation–spatial features of LiDAR data. The operations of the AFE module can be expressed as

$$U_H = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{ConvLSTM}\left(X_H\right)\right)\right) \tag{1}$$

$$U_L = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(X_L\right)\right)\right) \tag{2}$$

where BN and ReLU represent batch normalization (BN) and the ReLU activation function, respectively. ConvLSTM transforms X_H into attribute feature $U_H \in \mathbb{R}^{p\times p\times c}$. Conv is a convolutional layer with kernel size 3×3, which increases the channel number of the elevation feature from l to c, while the spatial size remains to be $p \times p$ and the output feature is defined as $U_L \in \mathbb{R}^{p\times p\times c}$.

2) MSC Module: MSCs enable the capture of features at various scales in HSI and LiDAR data. Small kernel size focuses on features within small regions, effectively extracting local features including spectral details of HSI, contour details, and texture features of the ground in LiDAR data. Large kernel size can capture global features, including a wide range of surface morphology and spatial structure information from HSI and LiDAR. Integrating these multiscale features results in richer feature representations. Our proposed MSC module, illustrated in Fig. 3, innovatively employs multiple parallel branches to extract specific scale features. These features are concatenated and fused with information from other branches several times, resulting in stable global–local features that enhance the representation of RS image structure. In addition, by controlling the number of convolution kernels, the accumulation of redundant information can be avoided.

Take the HSI branch as an example, the attribute feature U_H is fed into the MSC module, and the whole process can be formulated as

$$X_k^1 = \mathrm{Conv}^{k\times k}\left(U_H\right) \quad k = 3, 5, 7 \tag{3}$$

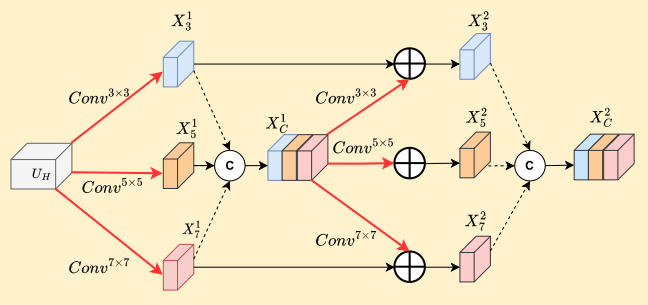


Fig. 3. Graphical illustration of MSC module.

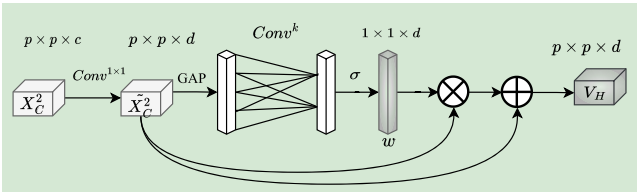


Fig. 4. DT module.

$$X_C^1 = \mathrm{Concat}\left(X_3^1, X_5^1, X_7^1\right) \tag{4}$$

$$X_k^2 = \mathrm{Conv}^{k\times k}\left(X_C^1\right) + X_k^1 \quad k = 3, 5, 7 \tag{5}$$

$$X_C^2 = \mathrm{Concat}\left(X_3^2, X_5^2, X_7^2\right) \tag{6}$$

where $\mathrm{Conv}^{k\times k}$ and Concat represent the convolutional layer with kernel size $k \times k$ and the concatenation operation, respectively. Each convolutional layer of the Conv operation is followed by a BN layer and an activation layer. $X_C^2 \in \mathbb{R}^{p\times p\times c}$ is the fused multiscale feature.

3) DT Module: In the previous modules, we obtained multiscale attribute features, which are the information source of common and unique features. To further embed the features into the subsequent transformer network, the extracted attribute features require tokenization. As is shown in Fig. 4, to linearly fuse the information of different channels, a convolutional

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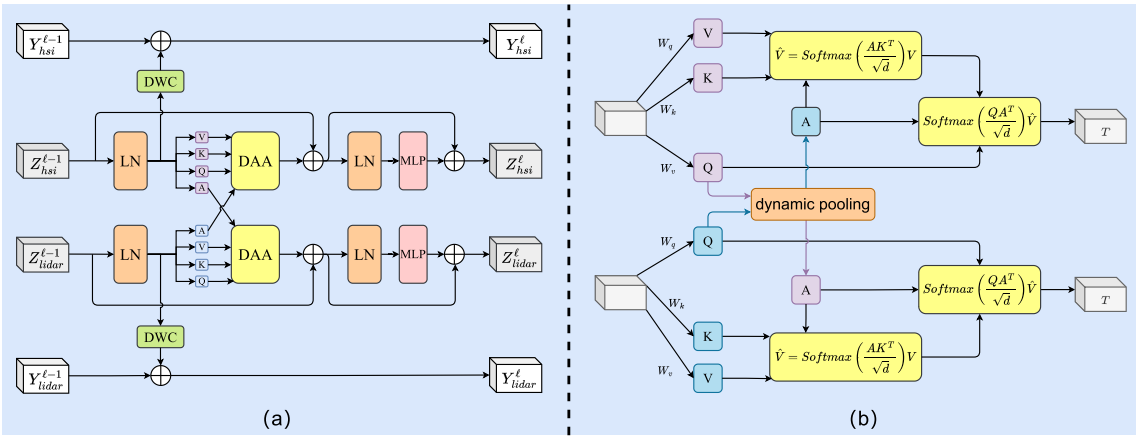


Fig. 5. (a) Graphical illustration of the ℓth CUT and (b) cross-DAA in the ℓth CUT. To facilitate intermodal information interaction to extract common features, the agent token A of the two modalities is exchanged, and then a DAA calculation is performed for each branch.

layer with kernel size 1×1 is employed to flexibly change the token dimension of X_C^2 from c to d. Then, channel attention is applied to reduce the impact of noisy bands while preserving critical information. For the HSI branch, V_H can be obtained by the following equations:

$$\tilde{X_C^2} = \text{Conv}^{1\times 1}\left(X_C^2\right) \tag{7}$$

$$w = \sigma\left\{\text{Conv}^k\left[\text{GAP}\left(\tilde{X_C^2}\right)\right]\right\} \tag{8}$$

$$V_H = w \otimes \tilde{X_C^2} + \tilde{X_C^2} \tag{9}$$

where $\sigma(\cdot)$ and GAP represent the sigmoid function and global average pooling, respectively. Conv^k represents a 1-D convolution of size k and the symbol $\otimes$ denotes element-wise multiplication at channel dimension. Similarly, the enhanced tokens of LiDAR data $V_L \in \mathbb{R}^{p\times p\times d}$ are obtained by inputting U_L into the MSC and DT module.

In this block, a ConvLSTM layer and a 2-D convolutional layer are employed to extract spectral-spatial features of HSI and spatial–elevation features of LiDAR data in a simple and effective manner, which is beneficial for subsequent blocks to extract common and unique features. Specifically, the MSC module integrates rich multiscale features to capture meaningful global–local information of HSI and LiDAR data, which effectively handles land objects of varying sizes and shapes in the same geographical environment. In addition, channel attention mechanisms are utilized to adaptively focus on features of interest, enhancing the model's ability to perceive the ground surface.

B. Common-Unique Transformer

In multimodal RS image processing, modeling the dependence between different modalities is crucial, as it is the key to mining their complementary information. HSI and LiDAR obtained from the same scene typically share common features, such as spatial information that reflects the edges and shapes of the terrain. In addition, each modality possesses unique features. HSI has high spectral resolution, with rich spectral bands providing detailed information on material composition. LiDAR data offer stronger penetration and rich structural information, which contain the geometric features of objects. To avoid information confusion during the feature fusion stage, it is essential to properly separate common and unique features. We propose a CUT, illustrated in Fig. 5(a), to effectively separate the common and unique features of HSI and LiDAR data.

1) Common Features Extraction: We develop a novel cross-dynamic-agent-attention (Cross-DAA) method to extract common features of HSI and LiDAR data, which aims to progressively obtain shared information through dynamic interaction between the features of HSI and LiDAR data.

Different from ternary MSA in the original transformer, DAA is a novel quaternary attention that consists of two softmax attention operations. Building up from ternary attention, we define new agent tokens $A \in \mathbb{R}^{n\times d}$, which are dynamically generated by Q. Then use Q, K, V, and A to calculate quaternion dynamic agent attention $\text{DAA}(Q, K, V, A)$. For input features $Z \in \mathbb{R}^{s\times s\times d}$, the whole process can be formulated as

$$Q = \text{ZW}_q, K = \text{ZW}_k, V = \text{ZW}_v \tag{10}$$

$$A = \text{MaxPooling}\,(Q) \tag{11}$$

$$\tilde{V} = \text{Softmax}\left(\frac{\text{AK}^T}{\sqrt{d}}\right)V \tag{12}$$

$$T = \text{Softmax}\left(\frac{\text{QA}^T}{\sqrt{d}}\right)\tilde{V} \tag{13}$$

where $\text{Softmax}(\cdot)$ and MaxPooling represent the softmax function and max pooling operation, respectively. $\tilde{V} \in \mathbb{R}^{n\times d}$ is the first softmax attention between A, K, and V. The output of $\text{DAA}(\cdot)$ is defined as $T \in \mathbb{R}^{N\times d}$, which is the second Softmax attention calculated by Q, A, and $\tilde{V}$.

As shown in Fig. 5, given the input features of the ℓth CUT block in HSI branch, we obtain queries Q_H^ℓ, keys K_H^ℓ, and values V_H^ℓ through linear mapping. The agent tokens A_H^ℓ are obtained by Q_H^ℓ through max pooling. By adjusting the stride of the pooling operation, we can dynamically generate agent tokens of different sizes. Specifically, in the ℓth CUT block, the shape of the agent tokens A_H^ℓ is $\ell \times \ell \times d$. As the depth increases, the spatial size of A_H^ℓ becomes progressively

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larger. Similarly, we obtain Q_L^ℓ, K_L^ℓ, V_L^ℓ, and A_L^ℓ in LiDAR branch. To better extract common information between the two modalities, the cross-attention between HSI and LiDAR is computed. Specifically, we exchange the agent tokens A_H^ℓ and A_L^ℓ for two branches to aggregate and propagate the interaction information between HSI and LiDAR. In HSI branch, treat the agent tokens A_L^ℓ of LiDAR as queries and perform attention calculations between A_L^ℓ, Q_H^ℓ, K_H^ℓ, and V_L^ℓ to extract the features of interest to the LiDAR agent from HSI features. Similarly, in LiDAR branch, features of interest to the HSI agent from LiDAR features are obtained. As the depth increases, the degree of interaction in cross-DAA continues to grow, allowing for the extraction of more common features. The operations above can be expressed as follows:

$$T_{\text{hsi}}^\ell = \text{DAA}\left(Q_H^\ell, K_H^\ell, V_H^\ell, A_L^\ell\right) \quad \ell = 1, \ldots, L \tag{14}$$

$$T_{\text{lidar}}^\ell = \text{DAA}\left(Q_L^\ell, K_L^\ell, V_L^\ell, A_H^\ell\right) \quad \ell = 1, \ldots, L \tag{15}$$

where T_{hsi}^ℓ and T_{lidar}^ℓ represent the cross-DAA of the HSI branch and LiDAR branch, respectively. Finally, we apply layer normalization (LN) and multilayer perceptron (MLP) to T_{hsi}^ℓ and T_{lidar}^ℓ, and thus obtain their common features Z_{hsi}^ℓ and Z_{lidar}^ℓ. The whole process can be written as follows:

$$Z_{\text{hsi}}^{\tilde{\ell}} = T_{\text{hsi}}^\ell + Z_{\text{hsi}}^{\ell-1} \quad \ell = 1, \ldots, L \tag{16}$$

$$Z_{\text{hsi}}^\ell = f_{AP}\left(\text{MLP}\left(\text{LN}\left(Z_{\text{hsi}}^{\tilde{\ell}}\right)\right) + Z_{\text{hsi}}^{\tilde{\ell}}\right) \quad \ell = 1, \ldots, L \tag{17}$$

$$Z_{\text{lidar}}^{\tilde{\ell}} = T_{\text{lidar}}^\ell + Z_{\text{lidar}}^{\ell-1} \quad \ell = 1, \ldots, L \tag{18}$$

$$Z_{\text{lidar}}^\ell = f_{AP}\left(\text{MLP}\left(\text{LN}\left(Z_{\text{lidar}}^{\tilde{\ell}}\right)\right) + Z_{\text{lidar}}^{\tilde{\ell}}\right) \quad \ell = 1, \ldots, L \tag{19}$$

where f_{AP} represents average pooling, which is applied to reduce the redundancy of common features.

2) Unique Features Extraction: Depthwise convolutions (DWCs) are employed to extract unique features of HSI and LiDAR data. DWC can extract features from each channel of HSI, capturing the details and relationships between bands in each band, increasing the diversity of spectral features. Meanwhile, extracting detailed multivariate spatial information from each channel of LiDAR helps to construct the surface morphology structure. In addition, DWC reduces computational complexity and improves generalization ability compared to traditional convolutions. The process of extracting unique features of HSI and LiDAR can be written as follows:

$$Y_{\text{hsi}}^\ell = \text{DWC}\left(\text{LN}\left(Z_{\text{hsi}}^{\ell-1}\right)\right) + Y_{\text{hsi}}^{\ell-1} \quad \ell = 1, \ldots, L \tag{20}$$

$$Y_{\text{lidar}}^\ell = \text{DWC}\left(\text{LN}\left(Z_{\text{lidar}}^{\ell-1}\right)\right) + Y_{\text{lidar}}^{\ell-1} \quad \ell = 1, \ldots, L \tag{21}$$

where DWC represents depthwise convolutions. Y_{hsi}^ℓ and Y_{lidar}^ℓ are unique features of HSI and LiDAR, respectively.

From common and unique feature extraction, the novel cross-DAA is proposed, where the new agent tokens A that are max-pooled from Q without any learnable parameters are defined, facilitating the aggregation and propagation of cross-modal information. By calculating the DAA between Q, K, V, and A from the other modality, the features of the two modalities dynamically interact. This interaction progressively captures more common features of HSI and LiDAR, reducing inconsistency and redundancy between them. Consequently, it helps establish consistent classification standards and improves classification reliability. In addition, we employ lightweight DWC to increase the diversity of the two modalities and effectively extract detailed spectral features from HSI and detailed morphology features of LiDAR. These unique features and complementary information provide the model with enhanced classification and discrimination capabilities.

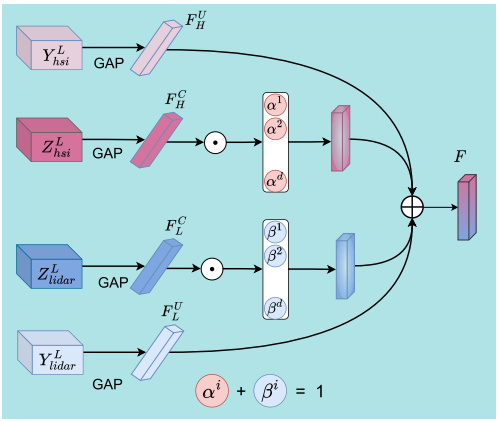


Fig. 6. Graphical illustration of CUFF block.

C. CUFF Block

To fuse the common and unique features of HSI and LiDAR data, the key is to enhance useful information while eliminating redundant information. As shown in Fig. 6, first, global average pooling is applied to eliminate information redundancy of common and unique features. Since there are two common features, they need to be fused into a single common feature. We define two learnable vectors $(\alpha^1, \ldots, \alpha^d)$ and $(\beta^1, \ldots, \beta^d)$, which map common features of HSI and LiDAR to the new features, respectively, and $\alpha^i + \beta^i = 1 (i = 1, \ldots, d)$. The fused common feature $F^C \in \mathbb{R}^{1\times1\times d}$ is obtained by combining the mapped features. Finally, F^C is added to the unique features of HSI and LiDAR to obtain the discriminative feature $F \in \mathbb{R}^{1\times1\times d}$. The whole process can be formulated as

$$F_H^C, F_L^C, F_H^U, F_L^U = \text{GAP}\left(Z_{\text{hsi}}^L, Z_{\text{lidar}}^L, Y_{\text{hsi}}^L, Y_{\text{lidar}}^L\right) \tag{22}$$

$$F^C = \alpha \odot F_H^C + \beta \odot F_L^C \tag{23}$$

$$F = F^C + F_H^U + F_L^U \tag{24}$$

where symbol $\odot$ denotes dot product. Integrating common and unique features helps to understand the contributions of different data sources and optimize feature selection. This fusion process ensures that the model leverages both common and unique information from HSI and LiDAR data, enhancing the interpretability and transparency of classification results.

Finally, F is fed into a classifier, with cross-entropy loss

$$\text{Loss} = -\sum y^i \log \hat{y}^i \tag{25}$$

where $\hat{y}^i$ is the predictive probability that the pixel belongs to class i. y^i is the expected probability that the pixel belongs to class i.

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TABLE I
CLASSES, TRAINING, AND TEST SAMPLE NUMBERS FOR HOUSTON 2013, TRENTO, MUUFL, AND AUGSBURG DATASETS

NO.	Houston 2013			Trento			MUUFL			Augsburg		
	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test
1	Healthy Grass	198	1053	Apple Trees	129	3905	Trees	150	23096	Forest	146	13361
2	Stressed grass	190	1064	Buildings	125	2778	Mostly Grass	150	4120	Residential	264	30065
3	Synthetic grass	192	505	Ground	105	374	Mixed Ground Surface	150	6732	Industrial Area	21	3830
4	Trees	188	1056	Woods	154	8969	Dirt and Sand	150	1676	Low Plants	248	26609
5	Soil	186	1056	Vineyard	184	10317	Road	150	6537	Allotment	52	523
6	Water	182	143	Roads	122	3052	Water	150	316	Commercial Area	7	1638
7	Residential	196	1072				Building Shadow	150	2083	Water	23	1507
8	Commercial	191	1053				Building	150	6090			
9	Road	193	1059				Sidewalk	150	1235			
10	Highway	191	1036				Yellow Curb	150	33			
11	Railway	181	1054				Cloth Panels	150	119			
12	Parking lot 1	192	1041									
13	Parking lot 2	184	285									
14	Tennis court	181	247									
15	Running track	187	473									
	Total	2832	12197	Total	819	29395	Total	1650	52037	Total	761	77533

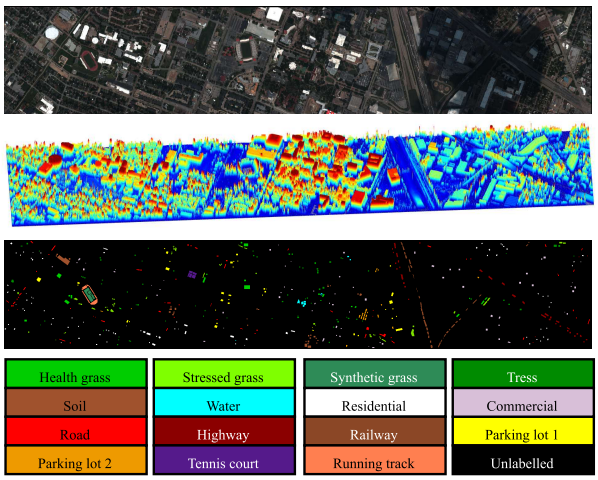


Fig. 7. Visual presentation of the Houston 2013 dataset. From top to bottom are the false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map, respectively.

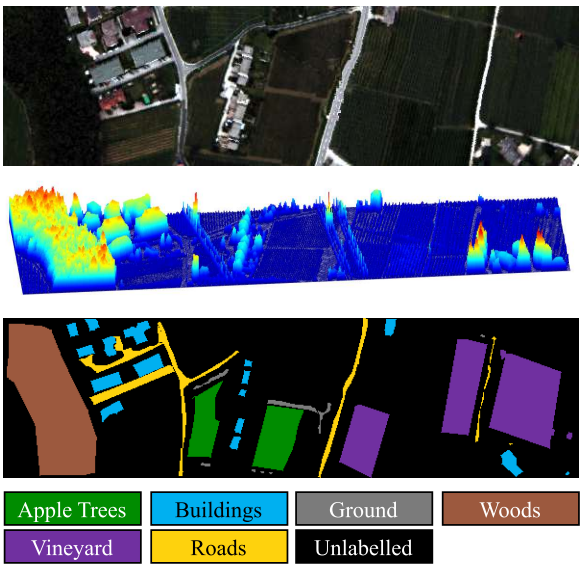


Fig. 8. Visual presentation of the Trento dataset. From top to bottom are the false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map, respectively.

IV. EXPERIMENT

We conducted experiments using four publicly available multimodal RS datasets to validate the effectiveness of the model. First, a brief description of the experimental data was provided. Next, the details of the experimental setup were described, and the experiment results were analyzed and compared with state-of-the-art algorithms to demonstrate the superiority of our method. Finally, we conducted ablation experiments to verify the effectiveness of each block.

A. Data Description

1) Houston Dataset [50]: The Houston dataset was acquired over the University of Houston campus and the neighboring urban area in June 2012. The dataset consists of an HSI and a LiDAR-derived DSM data, both of which contain 349 × 1905 pixels with a spatial resolution of 2.5 m, yielding 15 different classes in total. The numbers of samples for each class are reported in Table I. The HSI has 144 spectral bands with spectral resolution ranging from 380 to 1050 nm. The false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map are shown in Fig. 7.

2) Trento Data [30]: The Trento dataset was captured in a rural area in southern Trento, Italy, which contains HSI and one LiDAR data with 166 × 600 pixels and a spatial resolution of 1 m. The HSI has 63 spectral bands ranging from 380 to 1050 nm. The number of samples for six classes is reported in Table I. The false-color image of HSI, the 3-D LiDAR-derived DSM images, and the ground truth map are shown in Fig. 8.

3) MUUFL Dataset [51], [52]: The MUUFL dataset was collected in November 2010 at the University of Southern Mississippi's Gulfport campus in Long Beach, MS, USA. The HSI contains 64 spectral bands ranging from 380 to 1050 nm. Both HSI and LiDAR data contain 325 × 220 pixels with a spatial resolution of 1 m. The MUUFL dataset contains 11 classes. Table I lists the training and test sizes of each class. The false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map are shown in Fig. 9.

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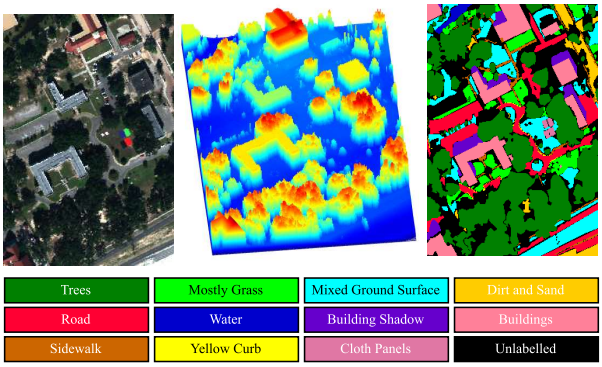


Fig. 9. Visual presentation of the MUUFL Gulfport dataset. From left to right are the false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map, respectively.

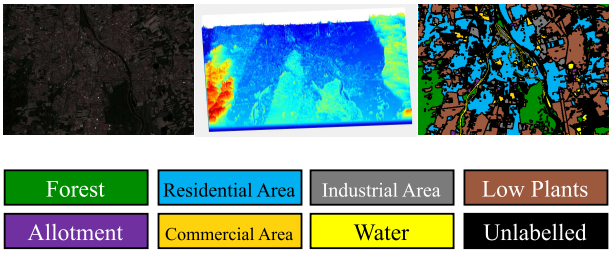


Fig. 10. Visual presentation of the Augsburg dataset. From left to right are the false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map, respectively.

4) Augsburg Dataset [53]: The Augsburg dataset consists of an HSI and a LiDAR image captured over the city of Augsburg, Germany. The HSI contains 180 spectral bands ranging from 400 to 2500 nm. Both HSI and LiDAR data contain 332×485 pixels with a spatial resolution of 30 m. The Augsburg data contain seven classes. Table I lists the training and test sizes of each class. The false-color image of HSI, the 3-D LiDAR-derived DSM image, and the ground truth label map are shown in Fig. 10.

B. Experimental Setup

Our proposed model is implemented in the Pytorch DL framework based on Python 3.6. The model is trained for 100 epochs with the Adam optimizer. The initial learning rate is set to 0.001 and decreases by 10% every ten epochs. All the experiments were run on an Intel Core i7-10700, 2.90-GHz processor, 64-GB RAM, and an NVIDIA RTX 3090 GPU.

Three metrics are used to measure the model performance, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). OA represents the proportion of correctly classified land cover to all samples, and AA represents the average classification accuracy of all classes. The range of Kappa (κ) values is $[-1, 1]$, which is used to measure the consistency of the data. Higher values of all three metrics indicate better classification results.

C. Parameter Setting

For the proposed DCU-Net, there are several parameters that need to be determined, i.e., the patch size p of input data, the dimension d of V_H and V_L, the attention heads h in cross-DAA, and the depth L of CUT. The detailed analysis is described below.

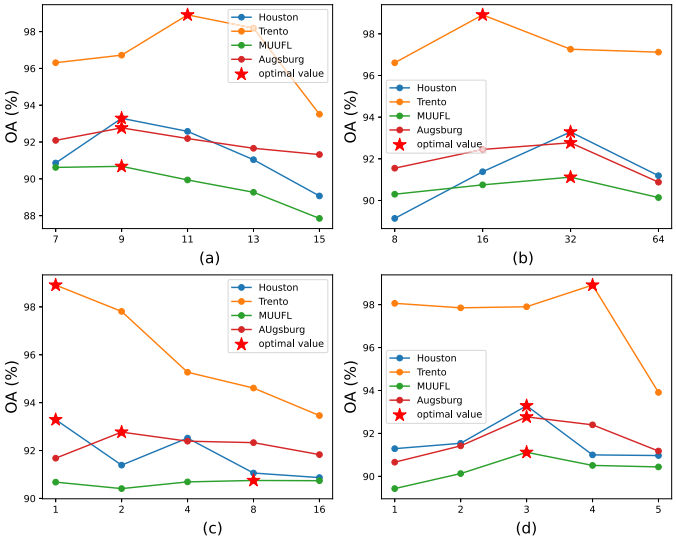


Fig. 11. OA (%) achieved by the proposed model with different parameters for the Houston 2013, Trento, MUUFL, and Augsburg datasets. (a) Patch size p. (b) Token dimension d. (c) Attention heads h. (d) Depth L.

1) Patch Size: To evaluate the effect of patch size p, we select p from {7, 9, 11, 13, 15} with other hyperparameters fixed. Fig. 11(a) shows the effect of different patch sizes p on the OA. When the patch size was set to 9, the optimal OA values were achieved in Houston, MUUFL, and Augsburg datasets, while the best p value for the Trento dataset is 11. This indicates that a moderate amount of spatial information is sufficient for our model to learn effective representations, and larger patches increase the redundancy of information.

2) Token Dimension: Similar to the analysis of patch size p, we fix the other hyperparameters and select d from {8, 16, 32, 64}. Fig. 11(b) shows how d value affects the classification performance on four datasets. The optimal d values for Houston 2013, Trento, MUUFL, and Augsburg datasets are 32, 16, 32, and 32, respectively. The reason is that spectral band of Trento and MUUFL is 63 and 64, respectively, while the HSIs of Houston and Augsburg have 144 and 180 spectral bands, respectively, which contain more spectral information. The optimal token dimension for each dataset depends on the spectral bands of each dataset.

3) Attention Heads: We test the optimal number of attention heads h for each dataset from a candidate set {1, 2, 4, 8, 16}. Fig. 11(c) reveals the relationship between different h values and the OA value for each dataset. It indicates that $h = 1$ is optimal for the Houston 2013 and Trento datasets, while $h = 8$ is optimal for MUUFL dataset, and $h = 2$ is optimal for Augsburg dataset. Increasing the number of heads h does not significantly enhance performance and could degrade the classification accuracy in some cases.

4) Depth of CUT: We test the performance of the proposed model with the depth number L from 1 to 5. Fig. 11(d) shows the influence of the depth on the OA of four datasets. We found that $L = 3$ is optimal for Houston 2013, MUUFL, and

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TABLE II
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON 2013 DATASET

Class No.	MLP	CNN methods			Attention methods				Common & Unique	
	EndNet	TBCNN	CCRNet	S^2ENet	MFT	GLT-Net	Fusion-HCT	ExViT	CDDFuse	Proposed
1	82.53±1.23	83.10±0.52	83.10±2.38	**92.97±1.20**	80.53±2.58	83.10±0.16	82.24±0.98	81.29±0.06	81.67±3.45	82.43±2.68
2	82.05±0.78	83.27±0.16	84.68±2.31	94.87±1.02	84.49±2.31	84.59±1.23	94.74±2.04	83.83±2.03	80.45±0.96	**98.97±1.00**
3	**100±0.00**	**100±0.00**	97.23±1.36	98.42±0.02	98.22±0.32	99.01±0.08	93.47±1.33	98.42±0.32	86.14±1.52	99.60±0.03
4	91.57±1.02	93.37±2.40	92.80±2.10	96.78±1.30	89.30±3.62	91.67±2.31	97.82±0.95	92.80±1.32	91.19±2.58	**99.43±0.06**
5	99.34±0.96	**100±0.00**	99.81±0.03	**100±0.00**	**100±0.00**	98.30±0.91	99.62±0.04	**100±0.00**	99.72±0.07	**100±0.00**
6	95.11±0.62	99.30±0.13	96.50±0.36	**100±0.00**	95.80±1.03	90.21±1.26	83.92±3.00	95.80±1.20	93.01±0.90	95.80±1.63
7	83.12±1.36	89.37±1.85	**94.12±1.04**	85.91±1.30	74.35±3.66	87.78±1.58	89.37±1.66	81.53±1.58	87.31±2.65	83.40±3.88
8	85.76±2.01	91.74±1.63	78.35±2.58	88.79±2.56	86.51±2.42	81.10±2.64	93.26±2.40	**93.54±1.98**	69.80±4.23	90.12±1.95
9	78.47±1.46	80.64±1.72	85.93±2.31	87.06±3.01	87.72±2.00	81.16±2.31	86.97±2.20	80.08±2.61	82.15±1.69	**95.09±1.09**
10	63.61±0.64	66.51±2.06	64.29±3.91	65.44±3.78	74.32±1.85	62.84±3.66	**87.07±2.38**	81.95±3.69	50.58±5.63	78.86±3.20
11	87.67±2.44	83.02±2.34	86.53±1.69	96.49±0.94	98.10±0.17	84.91±2.01	92.79±1.00	97.15±0.96	71.44±3.85	**98.39±0.33**
12	93.56±3.75	93.47±0.86	96.16±1.32	93.66±0.61	98.66±0.35	95.00±2.00	99.52±0.06	98.27±0.03	55.72±5.46	**99.90±0.02**
13	80.70±2.06	**92.28±0.65**	80.35±1.95	85.26±1.27	90.18±1.68	91.23±1.57	96.67±1.00	74.04±3.88	90.88±2.40	86.67±2.06
14	**100±0.00**	99.60±0.08	91.90±0.63	99.98±0.01	**100±0.00**	96.36±0.76	**100±0.00**	**100±0.00**	98.79±0.06	**100±0.00**
15	98.31±0.85	**100±0.00**	99.15±0.02	99.37±0.02	98.94±0.05	97.46±1.25	**100±0.00**	97.67±0.17	86.26±1.36	**100±0.00**
OA(%)	86.28±0.97	88.11±0.55	87.62±1.25	90.25±1.56	88.70±1.65	87.09±1.26	92.60±0.76	89.70±1.10	78.78±2.38	**93.29±0.22**
AA(%)	88.12±0.78	90.38±0.93	88.73±2.45	91.67±2.46	90.48±0.99	88.71±2.01	92.50±0.96	90.43±0.68	81.67±2.52	**93.91±0.19**
$\kappa\times100$	85.14±1.03	87.14±1.67	86.60±0.35	89.43±1.04	87.73±1.43	86.03±0.82	91.97±1.62	88.82±0.95	77.06±3.69	**92.71±0.24**

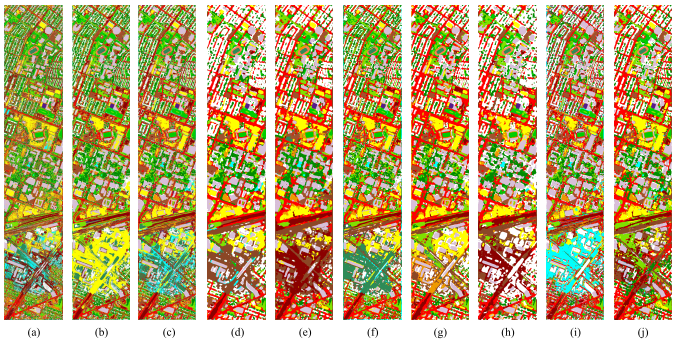


Fig. 12. Classification maps obtained by nine methods on the Houston 2013 dataset. (a) EndNet, (b) TBCNN, (c) CCRNet, (d) S^2ENet, (e) MFT, (f) GLC-Net, (g) Fusion-HCT, (h) ExViT, (i) CDDFuse, and (j) proposed.

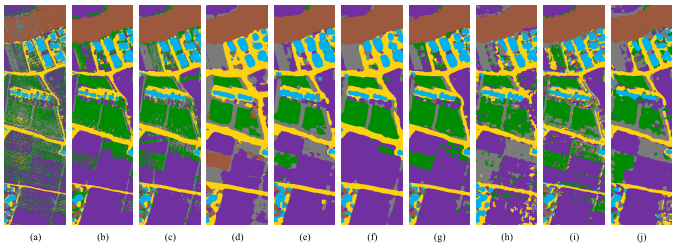


Fig. 13. Classification maps obtained by nine methods on the Trento dataset. (a) EndNet, (b) TBCNN, (c) CCRNet, (d) S^2ENet, (e) MFT, (f) GLC-Net, (g) Fusion-HCT, (h) ExViT, (i) CDDFuse, and (j) proposed.

Augsburg datasets, while $L = 4$ is optimal for Trento dataset. Therefore, we draw the conclusion that different datasets have varying common features between HSI and LiDAR data.

D. Classification Performance and Analysis

To verify the advancement and effectiveness of the proposed DCU-Net model, experiments are performed to compare it against several state-of-the-art classification methods, including EndNet [54], two-branch CNN (TBCNN) [30], cross-channel reconstruction network (CCRNet) [55], spatial–spectral enhancement network (S^2ENet) [56], MFT [45], global–local transformer network (GLT-Net) [43], fusion with hierarchical CNN and transformer (Fusion-HCT) [44], ExViT [46], and correlation-driven feature decomposition fusion (CDDFuse) network [49]. Among these methods, TBCNN, CCRNet, and S^2ENet are based on CNN architectures, while MFT, GLT-Net, Fusion-HCT, and ExViT are based on attention mechanisms. EndNet adopts fully connected neural networks. CDDFuse and our proposed DCU-Net are based on common and unique feature decomposition and fusion. To ensure fairness, all comparison methods share the same training set and testing set with our proposed DCU-Net model, and all compared methods use the default parameters in their original papers. Five training sessions are conducted and tested on each dataset to ensure accurate and stable results.

The OA, AA, κ, and preclass accuracy of the proposed model and the comparative methods for the Houston 2013, Trento, MUUFL, and Augsburg datasets are shown in Tables II–V, respectively. The classification maps of several comparison methods are shown in Figs. 12–15 for the four datasets, respectively.

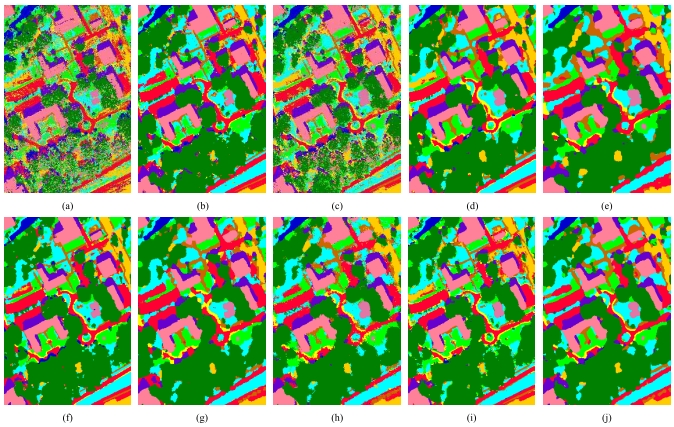


Fig. 14. Classification maps obtained by nine methods on the MUUFL dataset. (a) EndNet, (b) TBCNN, (c) CCRNet, (d) S^2ENet, (e) MFT, (f) GLC-Net, (g) Fusion-HCT, (h) ExViT, (i) CDDFuse, and (j) proposed.

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TABLE III

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET

Class No.	MLP	CNN methods			Attention methods				Common & Unique	
	EndNet	TBCNN	CCRNet	S^2ENet	MFT	GLT-Net	Fusion-HCT	ExViT	CDDFuse	Proposed
1	89.60±3.39	99.15±0.52	**100±0.00**	94.62±1.20	94.72±0.25	98.18±0.01	99.03±0.61	94.01±1.20	95.47±1.35	97.29±1.30
2	96.80±0.43	99.28±0.43	97.98±0.96	99.42±0.15	97.37±1.23	98.92±0.03	**99.89±0.10**	94.60±2.30	99.60±0.06	97.95±0.98
3	93.85±2.20	99.73±0.18	81.55±2.36	95.72±0.99	**100±0.00**	72.46±5.21	93.58±1.30	90.91±1.52	93.58±2.01	**100±0.00**
4	99.21±0.16	97.90±2.17	99.99±0.01	**100±0.00**	**100±0.00**	**100±0.00**	99.80±0.10	99.58±0.04	99.88±0.03	99.83±0.06
5	79.24±4.98	99.73±0.23	95.19±1.00	99.78±0.04	**99.92±0.02**	99.85±0.06	99.64±0.11	88.07±1.20	91.26±1.42	99.87±0.11
6	85.98±3.60	86.07±3.60	85.98±2.03	92.92±1.20	87.78±2.31	89.74±1.24	87.81±2.67	94.20±1.68	82.24±2.86	**95.80±1.69**
OA(%)	89.56±1.18	97.64±0.18	96.43±1.25	98.36±0.28	97.75±1.40	98.19±0.01	98.33±1.33	93.66±0.66	94.33±0.98	**98.91±0.13**
AA(%)	91.28±0.40	96.98±0.39	93.45±0.96	97.08±1.03	96.63±1.10	93.19±0.23	96.63±0.63	93.56±1.56	93.67±1.67	**98.46±0.51**
$\kappa\times100$	86.25±1.47	96.83±0.24	95.24±1.35	97.81±0.38	96.99±0.52	97.57±0.18	97.76±0.59	91.63±1.60	92.47±0.23	**98.55±0.18**

TABLE IV

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET

Class No.	MLP	CNN methods			Attention methods				Common & Unique	
	EndNet	TBCNN	CCRNet	S^2ENet	MFT	GLT-Net	Fusion-HCT	ExViT	CDDFuse	Proposed
1	88.21±1.25	91.61±2.03	91.16±1.36	91.77±1.20	89.00±1.00	**94.52±1.04**	92.47±1.21	91.41±0.94	93.90±1.90	92.74±1.68
2	87.21±1.05	80.87±3.05	80.32±1.66	86.41±1.14	84.44±1.44	86.43±1.65	83.52±2.06	83.81±2.18	80.00±2.08	**87.65±1.29**
3	70.00±3.26	73.17±4.17	73.23±3.21	**87.55±1.25**	77.97±3.09	72.88±2.78	83.59±1.36	77.29±3.62	80.14±2.88	85.50±2.49
4	91.35±2.71	89.02±2.04	92.01±1.24	95.41±0.41	96.18±0.98	**98.75±0.96**	93.85±1.44	95.23±1.23	90.04±1.20	96.96±1.67
5	**91.27±1.03**	89.97±1.06	87.38±1.69	85.91±0.19	90.55±2.00	86.48±1.46	89.72±1.27	83.97±2.97	86.92±3.12	88.99±1.83
6	98.42±0.63	**100±0.00**	**100±0.00**	99.68±0.03	**100±0.00**	**100±0.00**	99.05±0.25	**100±0.00**	**100±0.00**	99.68±0.21
7	87.66±1.66	91.31±2.55	92.90±1.20	**94.14±0.41**	91.69±0.98	93.42±0.93	90.69±1.39	89.77±2.17	89.92±2.63	93.33±1.33
8	92.51±1.20	96.06±1.85	96.85±1.30	95.65±0.65	92.99±1.32	**97.14±1.41**	94.73±1.37	96.57±1.20	97.13±0.96	95.45±1.58
9	77.57±3.75	76.36±2.96	75.87±2.71	78.14±2.41	66.64±4.08	60.65±3.56	72.96±1.53	67.21±2.58	77.17±2.98	**78.70±1.28**
10	**100±0.00**	81.82±2.54	90.91±0.91	84.85±1.58	66.67±3.66	81.26±1.06	78.79±2.09	87.88±2.81	93.94±0.40	81.82±1.83
11	**100±0.00**	99.16±0.44	99.16±0.16	99.16±0.34	99.16±0.62	99.16±0.62	99.16±0.62	99.16±0.60	95.80±0.98	99.16±0.60
OA(%)	86.59±1.03	88.30±0.35	87.98±1.04	90.47±0.37	87.76±0.20	89.70±0.12	90.09±0.25	88.20±1.20	89.88±0.98	**91.12±0.16**
AA(%)	89.47±0.80	88.12±1.21	89.07±1.28	90.79±1.93	86.84±0.42	88.29±0.52	88.96±0.29	88.39±1.09	89.54±0.14	**90.91±0.19**
$\kappa\times100$	82.50±1.40	84.63±0.12	84.24±0.96	87.48±0.25	84.03±1.71	86.4o±0.14	86.96±0.33	84.49±1.28	86.64±0.69	**88.31±0.24**

TABLE V

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET

Class No.	MLP	CNN methods			Attention methods				Common & Unique	
	EndNet	TBCNN	CCRNet	S^2ENet	MFT	GLT-Net	Fusion-HCT	ExViT	CDDFuse	Proposed
1	81.42±1.20	94.90±0.68	99.27±0.05	97.07±0.86	93.11±1.24	**99.33±0.35**	93.16±1.60	94.87±0.78	94.18±1.20	96.59±0.46
2	80.65±1.65	97.19±0.95	98.29±0.02	98.04±0.09	96.19±1.33	**98.63±0.04**	98.27±0.41	96.03±0.57	90.62±0.26	98.27±0.08
3	00.70±0.02	**76.95±1.03**	04.54±0.02	58.20±3.04	48.43±2.45	06.08±0.16	53.86±2.49	25.61±1.47	20.00±1.05	66.06±2.32
4	81.83±1.30	83.39±2.01	93.96±0.82	96.21±1.16	90.06±0.96	95.49±0.69	91.70±0.98	92.60±1.42	87.28±0.86	**97.64±0.06**
5	03.44±0.01	50.10±0.98	65.58±1.23	45.32±3.45	60.99±1.88	**68.83±1.86**	52.39±2.44	66.92±1.09	32.89±2.22	58.32±2.30
6	00.00±0.00	00.00±0.00	00.00±0.00	05.07±0.16	**15.69±1.20**	00.00±0.00	12.09±0.83	01.77±0.02	00.00±0.00	2.63±0.03
7	29.86±2.03	28.14±0.85	32.18±1.14	32.32±2.33	**48.11±2.41**	41.34±2.41	46.58±2.13	24.42±3.49	15.13±2.67	40.81±2.55
OA(%)	74.03±1.03	87.34±0.96	88.76±1.60	91.68±1.02	88.32±0.82	89.70±0.97	89.81±0.31	87.59±1.09	83.09±2.44	**92.77±0.68**
AA(%)	39.70±0.61	61.52±1.69	56.26±2.41	61.75±2.01	64.66±1.63	58.53±1.77	64.01±1.47	57.46±3.66	48.59±2.83	**65.76±1.86**
$\kappa\times100$	61.60±1.26	81.78±0.82	83.44±1.58	87.94±0.92	83.33±0.33	84.82±1.24	85.36±0.92	82.04±1.88	74.97±1.97	**89.52±1.07**

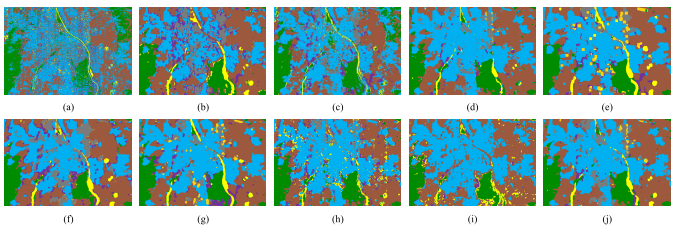


Fig. 15. Classification maps obtained by nine methods on the Augsburg dataset. (a) EndNet, (b) TBCNN, (c) CCRNet, (d) S^2ENet, (e) MFT, (f) GLC-Net, (g) Fusion-HCT, (h) ExViT, (i) CDDFuse, and (j) proposed.

1) Result on the Houston 2013 Dataset: As shown in Table II, the OA, AA, and κ of the proposed model on Houston data are 93.29%, 93.91%, and 92.71%, respectively, which are significantly better than other methods. The OA improvements achieved by proposed DCU-Net are 14.51% compared with CDDFuse. CDDFuse primarily extracts high-frequency information as unique features and low-frequency information as common features from multimodal data. However, this approach has inherent limitations. In contrast, our proposed DCU-Net incorporates the CUT block, which adaptively extracts both common and unique features from multimodal data, resulting in a more robust and effective performance. In terms of classification accuracy, the proposed model achieves the highest accuracy in class 1 "Stress grass," class 2 "Trees," class 5 "Soil," class 9 "Road," class 11 "Railway," class 12 "Parking lot 1," class 14 "Tennis court," and class 15 "Running track." Notably, the accuracies of "Soil," "Tennis Court," and "Running track" reach 100%. From the classification map shown in Fig. 12, it can be seen that our proposed model exhibits superior classification

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TABLE VI
COMPARISON OF MODEL COMPLEXITY AND TRAINING/TESTING TIME FOR DIFFERENT METHODS ON THE HOUSTON 2013 DATASET

Methods	MLP	CNN methods			Attention methods				Common & Unique	
	EndNet	TBCNN	CCRNet	S^2ENet	MFT	GLT-Net	Fusion-HCT	ExViT	CDDFuse	Proposed
#param (K)	91.97	105.90	76.15	270.87	313.07	545.63	728.35	229.10	1247.30	344.95
MFLOPs	0.09	1.80	0.53	262.13	202.44	312.28	71.20	366.74	148.42	355.53
training time (s)	0.1082	0.4289	0.6683	0.5037	2.1146	6.1207	2.0388	1.9278	11.8865	5.2636
testing time (s)	0.0329	1.1618	0.2144	3.3332	4.5681	5.6243	4.7996	4.8126	1.6998	4.1671

performance compared to other methods, with clearer boundaries and more accurate delineation of areas under cloud cover, which is consistent with the quantitative results.

2) Result on the Trento Dataset: As shown in Table III, the OA on several methods for the Trento dataset exceeds 98%, including S^2ENet, GLT-Net, and Fusion-HCT. Our proposed model has improved the OA by 0.55%, 0.72%, and 0.58% compared to these methods, respectively. As shown in Fig. 13, it can be seen from the classification map that S^2ENet, MFT, GLT-Net, Fusion-HCT, and our model have better classification performance with fewer confusion areas. The classification map of EndNet has a lot of noise because its input is pixels instead of patches, resulting in a lack of spatial information.

3) Result on the MUUFL Dataset: MUUFL dataset is a challenging dataset due to its smaller size of training samples and larger size of test samples. However, our proposed model achieves OA, AA, and κ of 91.12%, 90.91%, and 88.31%, respectively, outperforming other methods. From the classification map in Fig. 14, it can be seen that the proposed model can correctly classify the building area and vegetation area, which produces a clearer classification map with less noise.

4) Result on the Augsburg Dataset: Table V illustrates the classification performance of different methods on the Augsburg dataset. The OA, AA, and κ of the proposed model are 92.77%, 65.76%, and 89.52%, respectively, which are better than other methods. Due to the limited number of training samples in the class 6 "Commercial Area," the classification performance of all methods in class 6 suffers. The classification map shown in Fig. 14 demonstrates that the proposed DCU-Net can correctly classify the industrial and residential area, which is a challenging task for the other methods.

E. Computational Complexity Analysis

Table VI shows the model complexity of different methods on the Houston 2013 dataset, with all results representing the averages from five training sessions and tests. Among these methods, CCRNet has the smallest number of parameters, 76.15k, while CDDFuse has the largest, with 1247.30k parameters. In terms of computational complexity, EndNet is the most efficient as it inputs pixels instead of patches, whereas ExViT exhibits the highest computational demand, with an MFLOPs count of 366.74. The proposed DCU-Net strikes a balance between these extremes, with 344.95k parameters and a computational complexity of 355.53 MFLOPs. The main computational cost of DCU-Net lies in the cross-DAA of CUT block, which enhances classification results while keeping the increase in the number of parameters acceptable. In addition, we evaluate the training and testing time of different methods on the Houston 2013 dataset. It can be observed that the MLP and CNN methods achieved shorter training and testing time. The proposed DCU-Net demonstrates a training time of 5.2636 s and a testing time of 4.1671 s, both of which fall within an acceptable range.

TABLE VII
ABLATION ANALYSIS OF ESSENTIAL COMPONENTS

Models	Baseline	DT	CUT	CUFF	OA(%)			
					Houston 2013	Trento	MUUFL	Augsburg
0	✓				88.70±0.93	94.98±1.35	87.33±0.89	87.92±0.96
1	✓		✓		91.61±0.15	97.18±2.21	89.58±0.12	90.16±1.13
2	✓	✓	✓		91.99±0.57	97.98±1.26	90.45±0.38	90.69±0.95
3	✓		✓	✓	92.58±1.23	98.36±0.98	90.82±0.19	91.98±0.06
Proposed	✓	✓	✓	✓	**93.29±0.22**	**98.91±0.13**	**91.12±0.16**	**92.77±0.68**

F. Ablation Studies

1) Ablation Analysis of Essential Components: We conduct ablation experiments on four datasets to analyze the impact of different components on joint classification performance. Initially, we remove the proposed DT module, CUT block, and CUFF block from DCU-Net to get the base accuracy. As listed in Table VII, the baseline achieves 88.70%, 94.98%, 87.33%, and 87.92% OA (standard deviation = 0.93%, 1.35%, 0.89%, and 0.96%) for the Houston 2013, Trento, MUUFL, and Augsburg datasets, respectively. When employing only CUT block, model 1 establishes the dependence between HSI and LiDAR through common and unique features, and the OA values on the four datasets increase by 2.91%, 2.2%, 2.25%, and 2.24%, respectively. In model 2, we add the DT module to further embed the features into the transformer network, resulting in further improvements in classification performance. This illustrates that the tokenization process by the DT component contributes to the performance of the proposed framework. In model 3, we use both CUT and CUFF blocks; the OA values on the four datasets are improved compared with model 1. This demonstrates that CUFF can effectively fuse common and unique features and reconstructs them into discriminant features. When simultaneously using DT, CUT, and CUFF, the performance on the four datasets achieves the best.

2) Ablation Analysis of Cross-DAA: This section evaluates the design logic of the proposed cross-DAA. The results are

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TABLE VIII
ABLATION ANALYSIS OF CROSS-DAA

Attention Variants	OA(%)			
	Houston	Trento	MUUFL	Augsburg
MSA	89.10±0.66	95.58±2.01	89.45±0.36	89.92±0.15
static	90.10±0.09	90.49±1.45	90.09±0.15	90.23±1.06
Without Cross	90.78±0.75	95.23±0.63	89.96±1.68	90.39±0.46
proposed cross-DAA	**93.29±0.22**	**98.91±0.13**	**91.12±0.16**	**92.77±0.68**

TABLE IX
ABLATION ANALYSIS OF DIFFERENT MODAL DATA INPUT

Cases	OA(%)			
	Houston	Trento	MUUFL	Augsburg
HSI only	88.18±0.30	95.19±0.96	87.36±2.31	87.98±0.96
LiDAR only	67.83±1.53	86.58±2.38	53.96±3.45	71.28±3.12
HSI + LiDAR	**93.29±0.22**	**98.91±0.13**	**91.12±0.16**	**92.77±0.68**

presented in Table VIII. Replacing the first variant DAA with MSA results in decreased performance across all four datasets. In the second variant, we employ static agent attention, where the spatial size of agent tokens A_H^ℓ and A_L^ℓ does not vary with the depth of CUT and remains to be 3×3. This led to a significant decline in classification accuracy, including an 8.42% drop in OA on the Trento dataset. This illustrates that dynamic interaction between HSI and LiDAR can extract more refined common features compared to static interaction. In the third variant, we remove the cross operation and do not exchange the agent tokens between HSI and LiDAR when calculating attention. The experimental results confirmed the importance of cross-attention in multimodal tasks.

3) Ablation Analysis of Different Modal Data Inputs: We conduct ablation studies on four datasets to assess the impact of different modal data on classification performance, as shown in Table IX. Using only HSI as input achieves OA values of 88.18%, 95.19%, 87.36%, and 87.98% for the Houston 2013, Trento, MUUFL, and Augsburg datasets, respectively. In contrast, using only LiDAR data significantly reduces the performance, highlighting the importance of fusing HSI and LiDAR data. Our proposed DCU-Net effectively leverages the shared and complementary information from multimodal data, enhancing classification performance.

V. CONCLUSION

In this article, we propose a dynamic CUFF network for the joint classification of HSI and LiDAR data and mine the complex correlations between multimodal data. Specifically, our multiscale attribute feature extraction block effectively captures remarkable spatial–spectral information of HSI and spatial–elevation information of LiDAR. Furthermore, we develop a novel CUT block with cross dynamic-agent-attention to extract common features of HSI and LiDAR data, as well as depthwise convolution modules focusing on their unique features. This approach models the dependence relationship between HSI and LiDAR data, enabling the full utilization of their shared and complementary information. In the fusion stage, common and unique features are adaptively reconstructed into discriminative features containing high-level semantic information. Extensive experiments on four popular HSI and LiDAR datasets illustrate the superiority and effectiveness of our proposed model, showcasing the great potential for multimodal RS data analysis.

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